CO4 AT3 INDUSTRY PROBLEM SOLVING TASK

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from datetime import datetime

# Question class

class Question:

    def __init__(self, question, options, answer):

        self.question = question

        self.options = options

        self.answer = answer

    def display(self):

        print(self.question)

        for option in self.options:

            print(option)

# Quiz class

class Quiz:

    def __init__(self, title):

        self.title = title

        self.questions = []

    def add_question(self, question):
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        self.questions.append(question)

    def evaluate(self, responses):
        score = 0
        for i in range(len(self.questions)):
            if responses[i] == self.questions[i].answer:
                score += 1
        return score

# StudentResponse class
class StudentResponse:
    def __init__(self, name):
        self.name = name
        self.responses = []

    def add_response(self, response):
        self.responses.append(response)

# Create Quiz
quiz = Quiz("Python Programming Quiz")

# Add questions
quiz.add_question(Question(
    "1. Which language is Python?",
    ["A. Programming Language",
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     "B. Database",
     "C. Operating System",
     "D. Browser"],
    "A"
))
quiz.add_question(Question(
    "2. Which symbol is used for comments in Python?",
    ["A. //",
     "B. #",
     "C. /* */",
     "D. --"],
    "B"
))
quiz.add_question(Question(
    "3. Which keyword is used to define a function?",
    ["A. function",
     "B. define",
     "C. def",
     "D. fun"],
    "C"
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))
quiz.add_question(Question(
    "4. Which data type stores multiple values?",
    ["A. int",
     "B. float",
     "C. list",
     "D. bool"],
    "C"
))
quiz.add_question(Question(
    "5. Which function is used to get input from the user?",
    ["A. scan()",
     "B. input()",
     "C. read()",
     "D. get()"],
    "B"
))
# Store questions and answers in a file
with open("questions.txt", "w") as file:
    for q in quiz.questions:
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        file.write(q.question + "\n")
        for option in q.options:
            file.write(option + "\n")

        file.write("Correct Answer: " + q.answer + "\n")
        file.write("\n")
# Student name
name = input("Enter student name: ")
student = StudentResponse(name)
print("\n========== ONLINE QUIZ ==========")
print("Quiz:", quiz.title)
print("Student:", name)
print("==================================\n")
# Take answers
for q in quiz.questions:
    q.display()
    response = input("Enter your answer (A/B/C/D): ").strip().upper()
    # Handle invalid response
    if response not in ["A", "B", "C", "D"]:
        print("Invalid answer! Marked as unanswered.")
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        response = ""
    student.add_response(response)
    print()
# Calculate score
score = quiz.evaluate(student.responses)
total = len(quiz.questions)
percentage = (score / total) * 100
# Result
if percentage >= 50:
    result = "PASS"
else:
    result = "FAIL"
# Generate result report
with open("results.txt", "a") as file:
    file.write("========== QUIZ RESULT ==========\n")
    file.write("Student Name: " + name + "\n")
    file.write("Quiz: " + quiz.title + "\n")
    file.write("Score: " + str(score) + "/" + str(total) + "\n")
    file.write("Percentage: " + str(percentage) + "%\n")
    file.write("Result: " + result + "\n")
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    file.write("====================================\n\n")

# Maintain quiz attempt log
with open("quiz_log.txt", "a") as file:
    date_time = datetime.now().strftime("%Y-%m-%d %H:%M:%S")
    file.write(
        date_time
        + " | Student: "
        + name
        + " | Score: "
        + str(score)
        + "/"
        + str(total)
        + "\n"
    )

# Display final result
print("========== QUIZ RESULT ==========")
print("Student Name:", name)
print("Quiz:", quiz.title)
print("Score:", score, "/", total)
print("Percentage:", percentage, "%")
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print("Result:", result)

print("===================================")

print("\nResult saved to results.txt")

print("Questions saved to questions.txt")

print("Attempt logged to quiz_log.txt")
```

OUTPUT:

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Enter student name: SIVARANGJENII

========== ONLINE QUIZ ==========
Quiz: Python Programming Quiz
Student: SIVARANGJENII
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```
========== QUIZ RESULT ==========
Student Name: SIVARANGJENII
Quiz: Python Programming Quiz
Score: 4/5
Percentage: 80.0%
Result: PASS
==================================
```